

Maruf Hossain

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EDUCATION

Wilfrid Laurier University

B.S. Computer Science & Mathematics

Jan. 2023 – Present

Waterloo, ON, Canada

Coursework: Operating Systems, Computer Networks, Database Systems, Machine Learning, Optimization

Leadership: Operations (Laurier MSA), VP Marketing & Operations (Laurier Cricket Club), VP Internal (CaseHacks)

EXPERIENCE

Aippa Health (Stealth AI Startup)

VP of Engineering

Oct. 2025 – Present

Waterloo, ON, Canada

- Designed backend architecture for an **AI-powered healthcare platform** using **FastAPI** and **PostgreSQL**, implementing structured data ingestion, validation, and schema design to ensure reliable patient data handling
- Built **secure data ingestion, asynchronous processing, and real-time inference pipelines** using **Python** and REST APIs, improving system reliability and enabling consistent data flow across distributed services

Sony Interactive Entertainment

Software Engineering Intern (Build & Tools)

May 2025 – Aug. 2025

California, USA

- Designed and optimized **CI/CD pipelines** by restructuring workflows, implementing caching, and introducing validation checks, reducing build failure rates by **20–25%** and improving deployment reliability across internal services
- Automated build, test, and monitoring workflows using **Python** and **REST APIs**, eliminating **10–15 hours/week** of manual intervention and reducing mean time to resolution (MTTR) from **2 hours to under 1 hour**
- Analyzed build logs, execution metrics, and failure patterns to identify bottlenecks, reducing average build times from **20 min to 12–14 min (30–40% faster)** and improving debugging efficiency across engineering teams
- Implemented **parallelized test execution, artifact caching, and pipeline instrumentation**, increasing CI throughput by **2×**, reducing pipeline latency by **25–35%**, and improving visibility into build performance across high-throughput systems

Mitsubishi Motors Corporation

Software Engineering Intern, Connected Car Systems

Apr. 2024 – Jul. 2024

Markham, ON, Canada

- Developed telematics software in **Python** and **C++** to process **real-time vehicle telemetry data**, supporting **10K+ events/min** and improving data consistency by **20–25%** across vehicle-to-cloud communication systems
- Enhanced **validation and testing pipelines** by introducing automated checks and data verification layers, reducing data inconsistencies by **30%** and improving reliability across multiple deployment environments
- Analyzed and debugged **vehicle communication pipelines** using system-level tracing and log inspection, reducing fault detection time from **10 min to 6–7 min** and improving end-to-end data accuracy

Qalam Consulting

Chief Technology Officer

Oct. 2023 – Jul. 2024

Waterloo, ON, Canada

- Led engineering for **multiple client-facing applications**, owning system architecture, backend development, and deployment, delivering production systems supporting **100–300+ users** and reducing feature delivery time from **5 days to 3–4 days**
- Designed and scaled **backend services and REST APIs** using **Node.js**, improving API response times by **25–35%** and enabling reliable handling of concurrent user requests across deployed applications

PROJECTS

WLU MSA Website | [Next.js](#), [TypeScript](#), [Payload CMS](#), [PostgreSQL](#) [GitHub](#) [Live](#)

Jan. 2022 – Present

- Architected and maintained a **server-side rendered platform** serving **1,500+ users**, integrating **Payload CMS**, **PostgreSQL**, and **Stripe**, reducing manual event and content management overhead by **40%**
- Designed and built **CMS-driven APIs and data pipelines** for event publishing, registration, and content updates, enabling consistent data flow and reducing admin intervention across workflows
- Optimized **SSR data fetching, caching, and query performance**, improving page load times from **1.8s to 1.2–1.3s** and increasing search visibility across high-traffic pages

TECHNICAL SKILLS

Languages/Frameworks: C++, Go, Python, TypeScript, JavaScript, React, Next.js, Node.js, Express, FastAPI

Databases/Tools: PostgreSQL, Redis, Firebase Realtime DB, Docker, Kubernetes, AWS (EC2, S3), Vercel

Technologies/Concepts: REST APIs, Microservices, Distributed Systems, WebSockets, Caching, Concurrency